



Functions in C

A function is a block of code that performs a specific task and can be reused anywhere in the program.

(ऐसा task/work जो बार-बार repeat हो रहा है तो उस task को function बनाकर use करते हैं)

❖ User-defined Function

- A User defined function in C is a function that is define by the programmer to perform a specific task and can be called whenever needed in the program.

- **Syntax of User define function**

void hira()

```
{  
    printf("Hello World !");  
    Printf("\n I'm Hira);  
}
```

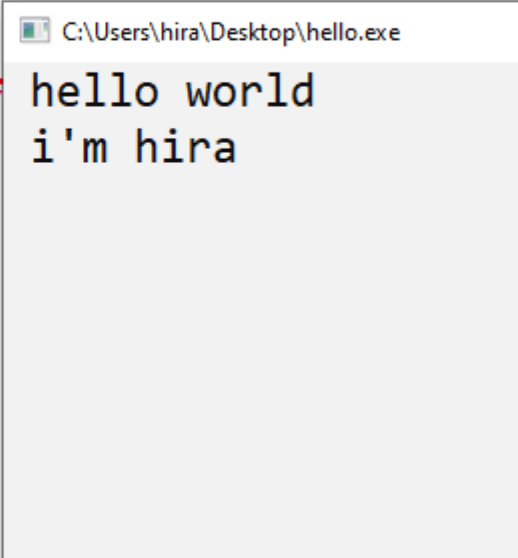
 // code to be executed

Here –

- ✓ hira() is the name of the function (User define).
- ✓ void means that the function does not have a return value.

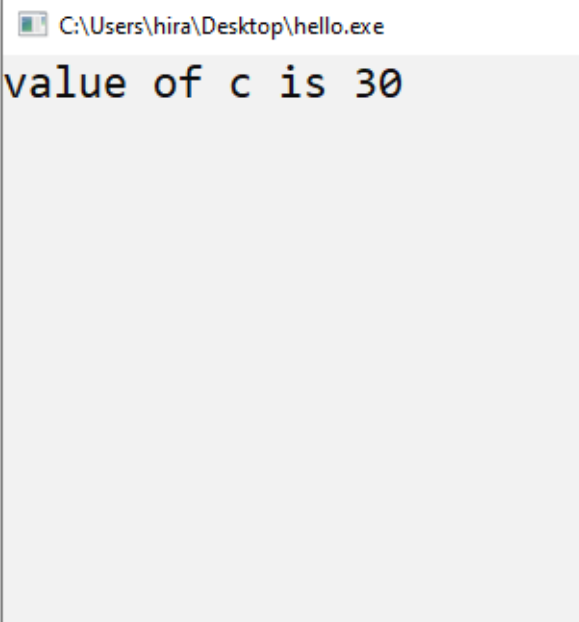
Q1. Print “hello world !” using function ?

```
#include<stdio.h>
#include<math.h>
void hira()
{
    printf(" hello world \n");
    printf(" i'm hira");
}
int main()
{
    hira();
    getch();
    return 0;
}
```



Q2. WAP to add two number using function without return value.

```
#include<stdio.h>
void hira()
{
    int a=10, b=20, c;
    c=a+b;
    printf("value of c is %d",c);
}
int main()
{
    hira();
    getch();
    return 0;
}
```



Q3. WAP to add two number using function with return value.

```

#include<stdio.h>
int hira()
{
    int a=10, b=20, c;
    c=a+b;
    return c;
}
int main()
{
    int a;
    a= hira();
    printf("value of c is %d",a);
    getch();
    return 0;
}

```

C:\Users\hira\Desktop\hello.exe
value of c is 30

❖ Function Declaration, Definition and call

- A **function declaration** tells the compiler about the function name, return type, and parameters before its actual definition.
It is also called the **function prototype**.
- A **function definition** contains the actual code that specifies what the function does.
- To **call a function**, write the function's name followed by two parentheses () and a semicolon ;

```

#include<stdio.h>
#include<conio.h>
void hira();
void hira()
{
    printf(" Welcome in Hello World ! Program ");
}
int main()
{
    printf("Hii, i'm Hira kumar \n");
    hira();
    getch();
    return 0;
}

```

Diagram illustrating the code structure:

- Function Declaration:** Indicated by an arrow pointing from `void hira();` to the text "Function Declaration".
- Function Definition:** Indicated by an arrow pointing from `void hira()` to the text "Function Definition".
- Function Call:** Indicated by an arrow pointing from `hira();` to the text "Function Call".

Feature	Function Declaration (घोषणा)	Function Definition (परिभाषा)
Purpose	Compiler को advance में बताना	Function का असली काम करना
Ends with semicolon	✅ हाँ	❌ नहीं
Body (Code Block)	❌ नहीं होता	✅ होता है

- ✓ Note: If a user-defined function, such as hira() is **declared** after the main() function, **an error will occur**. It is because C works from **top down approach**; which means that if the function is not declared above main(), the program is unaware of it:
- ✓ Error Occurs when the function is called before its declaration/prototype.
- ✓ Error occurs when function declared but not defined.